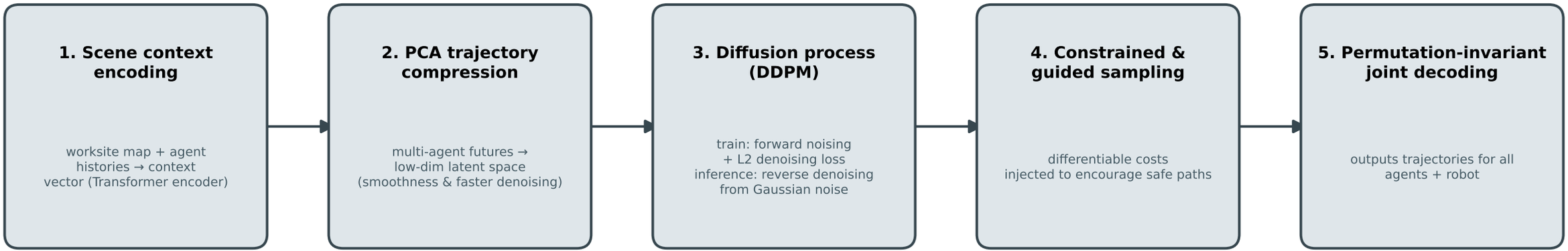
1. Scene context encoding
worksite map + agent histories → context vector (Transformer encoder)
2. PCA trajectory compression
multi-agent futures → low-dim latent space (smoothness & faster denoising)
3. Diffusion process (DDPM)
train: forward noising + L2 denoising loss
inference: reverse denoising from Gaussian noise
4. Constrained & guided sampling
differentiable costs injected to encourage safe paths
5. Permutation-invariant joint decoding
outputs trajectories for all agents + robot